

BRAC University

CSE331 : Automata And Computability
Spring 2025

Quiz 4 SET - A

Marks :20

Duration: 35 mins

CO3	Name:	ID:	Sec:
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1. Give a context-free grammar that generates the language
 - a. $L = \{ w \in \{a, b, c\}^* : w = a^i b^j c^k, \text{ where } j = 2i + k \text{ or } k = 2i + 2j, i, j, k \geq 0 \}$ [10]
 - b. $L = \{ w \in \{0,1\}^* : w \text{ contains even number of 0's} \}$ [4]
2. Convert the following regular expression into context free grammar. [6]
 $ab(ba + c)^* + ca^*b$

$$1) a) a^i b^{2i+k} c^k \\ = a^i b^{2i} b^k c^k$$

$$a^i b^j c^{2i+2j} \\ = a^i b^j c^{2j} c^{2i}$$

$$S \rightarrow A|P$$

$$A \rightarrow XY \\ X \rightarrow aXbb|\epsilon \\ Y \rightarrow bYc$$

$$P \rightarrow aPcc|Q \\ Q \rightarrow bQcc|\epsilon$$

$$b. S \rightarrow 0^* | 1S | \epsilon \\ X \rightarrow 1X | 0S$$

2.

$$ab(ba+c)^* + ca^*b$$

$$S \rightarrow XY$$

$$X \rightarrow abP$$

$$P \rightarrow QP|\epsilon$$

$$Q \rightarrow ba|c$$

$$Y \rightarrow CMb$$

$$M \rightarrow aM|\epsilon$$